AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Edged Chicago ORD01, IL -- station KDPA, America/Chicago
Building	OSM 1481581948 Edged Chicago ORD01
Facade gap	not applicable -- one building, no second facade
Weather record	43,457 real hourly records from KDPA
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-05-01 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 2 mode changes. The reactive incumbent took 14 hours -- 2 more -- but declared 2 unsafe hours against the agent's 0.

Agent	12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	14 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	4.715	21.0	4.440	1.002
01	FREE	yes	-	5.000	21.0	5.000	0.915
02	FREE	yes	-	4.165	21.0	3.330	0.842
03	FREE	yes	-	5.555	21.0	6.110	0.768
04	FREE	yes	-	6.390	21.0	7.220	0.717
05	FREE	yes	-	6.110	21.0	7.780	0.906

06	FREE	yes	-	8.080	21.0	8.330	0.892
07	FREE	yes	-	10.835	21.0	10.000	1.540
08	FREE	yes	-	13.610	21.0	12.220	2.078
09	FREE	yes	-	15.835	21.0	15.000	2.111
10	FREE	yes	-	17.225	21.0	16.670	1.831
11	FREE	yes	-	18.860	21.0	18.890	1.545
12	mechanical	no	dry-bulb	21.110	21.0	22.220	1.111
13	mechanical	no	dry-bulb	22.780	21.0	25.000	0.935
14	mechanical	no	dry-bulb	24.720	21.0	27.220	1.042
15	mechanical	no	dry-bulb	26.390	21.0	28.330	0.891
16	mechanical	no	dry-bulb	27.775	21.0	28.890	1.071
17	mechanical	no	dry-bulb	28.055	21.0	28.330	0.974
18	mechanical	no	dry-bulb	27.775	21.0	27.220	1.107
19	mechanical	no	dry-bulb	27.500	21.0	26.110	1.379
20	mechanical	no	dry-bulb	26.110	21.0	23.890	1.457
21	mechanical	no	dry-bulb	25.000	21.0	22.780	1.301
22	mechanical	no	dry-bulb	23.890	21.0	21.670	1.051
23	mechanical	no	dry-bulb	22.500	21.0	20.560	1.125

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.715 C, which is 16.285 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.002 C, and the margin is measured, not chosen: 1.002 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.000 C, which is 16.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.915 C, and the margin is measured, not chosen: 0.915 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.165 C, which is 16.835 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.842 C, and the margin is measured, not chosen: 0.842 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.555 C, which is 15.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.768 C, and the margin is measured, not chosen: 0.768 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.390 C, which is 14.610 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.717 C, and the margin is measured, not chosen: 0.717 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.110 C, which is 14.890 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.906 C, and the margin is measured, not chosen: 0.906 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.080 C, which is 12.920 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.892 C, and the margin is measured, not chosen: 0.892 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.835 C, which is 10.165 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.540 C, and the margin is measured, not chosen: 1.540 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.610 C, which is 7.390 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.078 C, and the margin is measured, not chosen: 2.078 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.835 C, which is 5.165 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.111 C, and the margin is measured, not chosen: 2.111 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.225 C, which is 3.775 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.831 C, and the margin is measured, not chosen: 1.831 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.860 C, which is 2.140 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.545 C, and the margin is measured, not chosen: 1.545 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.110 C against a 21.0 C limit, so it fails by 0.110 C. It would take a limit 0.110 C higher, or a bound 0.110 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.780 C against a 21.0 C limit, so it fails by 1.780 C. It would take a limit 1.780 C higher, or a bound 1.780 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.720 C against a 21.0 C limit, so it fails by 3.720 C. It would take a limit 3.720 C higher, or a bound 3.720 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.390 C against a 21.0 C limit, so it fails by 5.390 C. It would take a limit 5.390 C higher, or a bound 5.390 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.775 C against a 21.0 C limit, so it fails by 6.775 C. It would take a limit 6.775 C higher, or a bound 6.775 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.055 C against a 21.0 C limit, so it fails by 7.055 C. It would take a limit 7.055 C higher, or a bound 7.055 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.775 C against a 21.0 C limit, so it fails by 6.775 C. It would take a limit 6.775 C higher, or a bound 6.775 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.500 C against a 21.0 C limit, so it fails by 6.500 C. It would take a limit 6.500 C higher, or a bound 6.500 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.110 C against a 21.0 C limit, so it fails by 5.110 C. It would take a limit 5.110 C higher, or a bound 5.110 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.000 C against a 21.0 C limit, so it fails by 4.000 C. It would take a limit 4.000 C higher, or a bound 4.000 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.890 C against a 21.0 C limit, so it fails by 2.890 C. It would take a limit 2.890 C higher, or a bound 2.890 C tighter, to change this hour.

23:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.500 C against a 21.0 C limit, so it fails by 1.500 C. It would take a limit 1.500 C higher, or a bound 1.500 C tighter, to change this hour.

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